

# KULDEEP SINGH

## Software & System Engineering Lead | EV Controls, BMS and Vehicle Integration

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### PROFESSIONAL SUMMARY

Automotive software and systems engineering leader targeting roles in software engineering, system engineering, safer mobility, sustainable vehicle technologies, and future mobility solutions. Brings 8 years across EV/hybrid architecture, eVCU/BMS software, MATLAB/Simulink/Stateflow MBD, embedded C/C++, CAN diagnostics, HV battery systems, DCDC/OBC/PDU integration, vehicle validation, and quality-led development. Strong fit for roles needing cross-functional systems thinking, embedded software, EV powertrain integration, safety/process discipline, and customer-ready mobility products.

### TARGET ROLE FIT

- Software Engineering / System Engineering - Automotive
- EV Systems Integration Engineer
- Embedded Software Engineer - Mobility Systems
- BMS / Vehicle Controls Lead

### CORE STRENGTHS

- Software/system engineering fit: led EV architecture, ECU application software, component specifications, supplier interfaces, validation planning, and homologation readiness.
- Mobility technology fit: delivered EV and hybrid systems that improve efficiency, safety, and sustainability across commercial vehicle applications.
- Process fit: combined ISO 26262 functional safety awareness, IATF 16949 quality management, DFMEA, RCA, Six Sigma, and validation evidence into structured automotive delivery.

### PROFESSIONAL EXPERIENCE

#### Technical Manager - Product Development

IX Energy Pvt. Ltd., Noida, India | Jul 2018 - Present

- Led EV and P4 hybrid architecture, ECU application software, HV battery engineering, supplier development, validation, and certification readiness for commercial vehicle platforms.
- Developed MATLAB/Simulink/Stateflow eVCU/BMS logic for vehicle control, charging, DCDC/OBC/PDU coordination, fault handling, diagnostics, and safe operating states.
- Defined system and component specifications for motor, HV battery, PDU, DCDC, OBC, gearbox, controllers, EPS, BCS, TCS, HVAC, and embedded vehicle electronics.
- Managed cross-functional engineering and supplier teams to improve component readiness, validation efficiency, system integration, and release-quality software/hardware interfaces.
- Applied DFMEA, RCA, DMAIC, SPC, and control charts to reduce risk, improve supplier quality, and support automotive compliance deliverables.

### SELECTED PROGRAMMES & PROOF

- Commercial EV systems: LCV/LPT EV, 5T EV, sedan, and light commercial EV conversions with HV battery, charger, DCDC, controller, vehicle control, and validation integration.
- P4 hybrid bus: 16T ICAT-certified hybrid programme with super-capacitor pack and 25% fuel-efficiency improvement validation.
- In-house software/electronics: LV/HV PDU, STM32 instrument cluster, dual-CAN telematics, ultra-capacitor cell monitoring/control, and EV range estimation models.

### TECHNICAL TOOLKIT & CERTIFICATIONS

Software Engineering, System Engineering, MATLAB, Simulink, Stateflow, Embedded C/C++, BMS, eVCU, VCU, CAN, DBC, diagnostics, HV battery, DCDC, OBC, PDU, vehicle integration, ISO 26262, IATF 16949, DFMEA, RCA, Six Sigma.

- Six Sigma Black Belt - Certified
- ISO 26262 Functional Safety for Road Vehicles - TUV SUD Certified
- IATF 16949:2016 Automotive Quality Management - TUV SUD Certified
- AWS IoT & SIMULINK - Amazon Web Services / MathWorks Certified

### EDUCATION

B.Tech - Mechanical Engineering, VIT University, Vellore, Tamil Nadu | 2014 - 2018 | First Class, 75%